

Francesco FORCHER

PERSONAL DATA

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LANGUAGES: English C2, Italian native

ABOUT ME

Senior Software Systems Architect and **Machine Learning Engineer** with **over 8 years** of professional experience, driven by a lifelong passion for programming and sciences. **ETH Zürich CSE MSc**. Expertise includes delivering secure, proven, and performant software at all scales, **Rust, AI/Machine Learning, software architecture, distributed systems**, Computer Vision, robotics, physics and mathematics, and High-Performance Computing (HPC).

TECHNOLOGY STACKS

Currently: **Rust, Python, Kubernetes, AI/Machine Learning, Software Architecture and Design, Docker, Linux**
Previous experiences: **C++, Databases/(No)SQL, Julia, Javascript/Typescript, Java, R, Shell Scripting, Stochastic Differential Equations, Time Series, Matlab.**

EMPLOYMENT



Senior Software Systems Architect, Rust Consultant at **Thales Alenia Space**

DEC 2025 - CURRENT

Leading the development of a next-generation SDK, orchestration, and ML framework that enables end users to access and share on-board space edge-computing hardware platforms through an ultra-low-footprint, reliable, and secure integrated solution. Stack: **Rust, Kubernetes**, Go, Python.



Research Assistant at **Stockholm University**

APR 2024 - NOV 2025

Coauthor responsible for the AI research and ML engineering side of a research project in computer vision and political economy, led by professors Alex Lee and Per Andersson. We analyze the link between economic development and innovations in art and culture using multimodal AI embeddings to analyze large-scale art datasets efficiently, clustering image embeddings in latent space.



Database Developer at **Natzka**

JUN 2021 - MAR 2024

Contributed to the design and development of a next-generation **OLAP database for data warehousing**:

- Development of several **engine and query language features**, such as Boolean formulas and COUNT algorithms, with **Rust**. Contributed to **backend** development with **gRPC/REST** APIs, using **async, tokio, axum, serde** etc.
- Integration of AI/ML features into the product. Bringing Python and Julia scripts from research prototypes into production-ready **containerized microservices**, integrating state-of-the-art **Gaussian Process** based time series **forecasting services** optimized for **low latency**. Worked on a custom **Julia distributed logger** for **observability**.
- Research on a custom **Kubernetes predictive autoscaler**. Advanced load testing with **Grafana k6**.



Data Warehouse Consultant at **BLU software**

JAN 2021 - JUN 2021

- Integrated PostgreSQL and TimescaleDB as a data warehouse; used **Talend** and **Airflow** for ETL/ELT workloads and created interactive demand forecasts with Facebook's Prophet framework.



Data analyst, Physicist at **CERN, BE-ABP-HSS**

SEP 2016 - MAY 2018

CERN Technical Student in the Beams Department, Accelerator and Beam Physics Group.

- I developed advanced statistical **routines** using **Python Pandas** and **scikit-learn** to analyze crystal parameters for CERN's **UA9** experiment. My algorithm enabled the analysis of **very noisy** datasets, with a significant impact in the tender process to award the crystal manufacture contract worth **several hundred thousands CHF**.
- My **thesis** has been published as [CERN internal note](#).
- Using the C++ **R00T** library, I developed routines to analyze nuclear dechanneling for high-energy particle beams in bent crystals and improved the simulation accuracy of **SixTrack** software.

EDUCATION



ETH Zürich Master's Degree in COMPUTATIONAL SCIENCE AND ENGINEERING, Physics minor

SEP 2018 - DEC 2020

Master's degree with courses on high performance parallel computing, differential equations, Quantum Computing, Computational Statistics, Deep Learning and Computer Vision. I worked at **PSI** to develop an application to perform intrusive **Polynomial Chaos** expansion for **uncertainty quantification** in simulations based on approximate Hamiltonian maps, using **SymPy**.



University of Padova Bachelor's Degree in PHYSICS

OCT 2013 - SEP 2017

Physics degree with an emphasis in Computational Physics, simulations, statistics and data analysis.